MIL-T-60101A(MU)
18 July 1974
SUPERSEDING
MIL-T-60101(MU)
24 January 1964

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M9
ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Thruster, Cartridge Actuated, M9, Assembly loaded with Cartridge, Impulse, M119 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-P-19834	- Plates, Identification, Metal Foil, Adhesive Backed

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS28741	- Hose Assembly Detachable End Fitting, Medium Pressure

DRAWINGS
Armament Command

F10520478	- Thruster, Cartridge Actuated, M9 Assembly
C8845196	- Box, Fiberboard (Inner) Packing Ammunition for Thruster, C/A M9

FSC 1377

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2.1 (Cont'd)

DRAWINGS (Cont'd)

C8845197	- Box, Packing, Ammunition for Thruster, Cartridge Actuated M9
C9217287	- Box, Fiberboard (Outer) Alternative Packing, Ammunition for Thruster C/A, M9
IEL-10520478	- Inspection Equipment List for Thruster, Cartridge Actuated, M9 Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-Material, parts and assemblies shall comply with the requirements specified in the applicable drawings and specifications.

3.2 Torque.-Torque requirements shall be as specified in Table I. No movement shall occur upon application of the breakaway torque.

TABLE I

Component	A	B
	Seating or Bottoming (inch pounds)	Breakaway (inch pounds)
Breech & Body	200 ± 20	400 min.
Breech & Collar	200 ± 20	400 min.

3.3 Assembly

3.3.1 Trunnion rotation.-Upon completion of assembly the trunnion shall be free to rotate 360 degrees.

3.4 Residual magnetism.-The thruster shall n not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 Functional.

3.5.1 The thruster shall not fail to fire.

3.5.2 The thruster shall function without mechanical failure (see 6.1).

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3.5.3 Load.-The thruster fired under load shall perform as follows:

3.5.3.1 Propels a 50 pound mass horizontally against a resistive force of 800 pounds, minimum, for a stroke of 3.6 inches, minimum, to 3.8 inches maximum. (see 6.4)

3.5.3.2 Operating time.-The time between firing of the cartridge and completion of the stroke shall not exceed 80 milliseconds.

3.5.3.3 Bypass pressure.-The thrusters fixed under load shall furnish a bypass pressure of 1000 pounds per square inch (psi) minimum, at the end of 42 inches of aircraft hose, size 4, Standard MS28741, when the stroke has reached a minimum of 3.6 inches.

3.5.4 Ignition delay.-The ignition delay time shall not exceed 25 milliseconds (see 6.5)

3.5.5 Temperature.-The thruster shall comply with all the ballistic requirements throughout the temperature range of minus 65 degrees to plus 200 degrees F.

3.5.6 Initiation.-The thruster shall be fired by a source delivering an output equal to that of an M3A2 Initiator at the end of 15 feet of aircraft hose, size no. 4, MS28741, at 70 degrees $\pm$ 5 degrees F.

3.6 Workmanship.-Requirements for workmanship are as specified in the applicable drawings, referenced specifications and the following: 361 metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification and drawing where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.-Reference shall be made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with Specification MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.-The product shall be submitted in accordance with Standard MIL-STD-105.

4.2.1.2 Size of lot.A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.-One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table II. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. All nonconforming materials shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Impulse M119, Drawing D8595513, covering a detail of Drawing F10520478.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched or dented	Visual

4.2.2.2 Breech and Firing Pin, Sub-assembly consists of: Breech 10520483, Firing Pin 10520485, "O" Ring MS28775-009 covering a detail of Drawing F10520478.

CRITICAL	
1. Firing pin hole obstructed	Visual
MAJOR	
101. Thread damaged or not full	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly Drawing F10520478 less Trunnion 10520490 Collar 10520484 and Shipping Plug 8593299.

CRITICAL	
1. Head of breech in contact with body	Visual
MAJOR	
101. Excess sealant not removed	Visual
102. Thread damaged or not full	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
201. Protective finish inadequate or defective	Visual

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4.2.2.4 Assembly, Drawing F10520478, less Shipping Plugs 8593299.

Defect	Method of Inspection
CRITICAL	
1. Shear pin not engaged in firing pin	Gage
MAJOR	
101. Total length from center of trunnion to face of adapter	Gage
102. Total length	Gage
103. Thread damaged	Visual
104. Foreign matter in gas port	Visual
MINOR	
None defined	

4.2.2.5 Assembly, Drawing F10520478.

CRITICAL	
1. Any part missing, except as defined below	X-ray
2. Improper assembly, except as otherwise classified	X-ray
3. Shear pin missing in shear ring	X-ray
4. Cartridge missing	X-ray
5. Metal defective	X-ray
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Trunnion not free to rotate 360 degrees	Manual
102. Breech above flush	Visual
103. Threads damaged or not full	Visual
MINOR	
201. Total length max.	Measure
202. Protective finish defective	Visual
203. Shipping cap or plug missing	Visual
204. Location of labels, incorrect	Visual

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4.2.2.6 Box, Fiberboard (Inner) Drawing C8845196

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Box broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete or badly wrinkled	Visual

4.2.2.7 Box, Packing Drawing C8845197

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Strapping missing, loose, broken or improperly assembled	Manual/Visual
203. Contents of box exposed	Visual
204. Box damaged	Visual
205. Marking misleading or unidentifiable	Visual
206. Number of units incorrect	Visual

4.2.2.8 Box, Fiberboard (Outer) Alternative Packing Ammunition, Drawing C9217287.

CRITICAL	
None defined	
MAJOR	
None defined	

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4.2.2.8 (cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual/Visual
202. Carton broken or damaged	Visual
203. Number of units incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

4.2.3 Inspection testing. The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.

4.2.3.1.1 Breech and body.-Each breech and body sub-assembly shall be subjected to a breakaway torque application. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the sub-assembly.

4.2.3.1.2 Breech and collar.-Each breech and collar of the thruster assembly shall be subjected to a breakaway torque application. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the thruster.

4.2.3.2 Residual magnetism.-Each thruster assembly shall be tested for excessive residual magnetism. Failure of any thruster to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.3 Functional tests.-The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics.-Units shall be tested to determine compliance with the requirements of 3.5.1 and 3.5.8. The unit shall be temperature conditioned as provided in Table III and 4.3.4.1.1. Failure of any unit to comply with the requirements shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size	Method
-65 degrees to -70 degrees F	7	Load
200 degrees to 205 degrees F	7	Load

4.2.4 Inspection equipment.-The examination and tests shall be made using the equipment listed on IEL 10520478.

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4.3 Test methods and procedures.

4.3.1 Breakaway torque.

4.3.1.1 Breech and body.-The breakaway torque shall be applied to the breech after insertion of cartridge and the thread sealing compound has been cured in accordance with the drawing requirement. An air rise or other holding fixture shall be used to secure the body.

4.3.1.2 Breech and collar.-The breakaway torque shall be applied to the collar after the thread sealing compound has been cured in accordance with the drawing requirement.

4.3.2 Residual magnetism.-The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as the compass, in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly pass the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster around its longitudinal axis.

4.3.4 Functional.

4.3.4.1 Ballistics.-The test samples shall be rigidly supported so that the entire thrust is borne by the thruster mounting points. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures.

4.3.4.1.1 Temperature conditioning.- A simulated thruster and cartridge with a temperature sensitive element, placed in the dummy cartridge, shall be stored in a temperature conditioning chamber at the same time as the samples. The temperature conditioning time shall be a minimum of one hour longer than the time necessary for the temperature of the simulated cartridge to reach the required test temperature.

4.3.5 X-ray.-The equipment and procedure specified herein shall be used to perform the x-ray examination. The equipment shall be 80 Kilovolts, 10 milliamperes, 60 seconds. The target to film distance shall be 36 inches, minimum, using AA film. A high intensity viewer shall be used to determine deflectiveness of units.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity: Level A - In accordance with drawings C8845196 and C8845197; Level B - In accordance with drawings C8845196 and C9217287.

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6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Load.-The expected performance of the thruster under conditions simulating those existing in aircraft shall be referred to as load condition in this specification.

6.3 Ignition delay is defined as the time lapse between the firing of the cartridge and the beginning of the continuous pressure rise in the body of the thruster.

6.4 Explosive Hazard Classification Information:

Explosive Hazard Class - 1
Storage Compatibility Group - B, E, N
DOT Hazard Class - C
DOT Marking - Explosive Power Device, Class C
Handle Carefully - Keep Fire Away

6.5 Hazard notice.-The thruster described herein and certain of its components are flammable and/or explosive, consequently presents hazards in manufacture, handling, storage and shipments. The contractor should recognize these hazards to take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.6 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A584

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-60101A(MU) Thruster, Cartridge Actuated, M9 Assembly

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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